

# **PAPER\_1979 — Sombrero Dark Matter Mass Ratio $M_{DM}/M_{total} = 2 \cdot F_{TRZ} = 0.2$ EXACT as Candidate Cross-Domain Extension of PAPER\_1944's Magnetar Meissner-Boundary Identity Family — Narrow Attribution With Explicit Acknowledgment That the Underlying PAPER\_1944/1945 $n_{lobes} \cdot F_{TRZ}$ Framework May or May Not Extend to Dark Matter Physics**

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## **PAPER\_1979 — Sombrero $M_{DM}/M_{total} = 2 \cdot F_{TRZ}$ Cross-Domain Instance**

### **Abstract**

**PAPER\_1944 ("Magnetar  $B/B_{crit} = 2 \cdot F_{TRZ} = 0.2$  EXACT")** documents the seminal integer-primitive-lock identity:

``

For ALL magnetars whose UQFF classification is "below Meissner boundary":  
 $B_{surface} / B_{crit} = 2 \cdot F_{TRZ} = 0.2$  EXACT (universal primitive-lock)

``

anchored at SGR 1745-2900 (Chandra-measured  $B_{surface} = 2 \times 10^{11}$  T, PAPER\_266's  $B_{crit} = 10^{11}$  T UQFF Gravitational Meissner Boundary).

**PAPER\_1945 ("Magnetar  $n_{lobes} \times F_{TRZ}$  Universality Confirmed")** generalizes PAPER\_1944's identity to:

``

For any below-Meissner-boundary magnetar:  
 $B_{surface} / B_{crit} = n_{lobes} \cdot F_{TRZ}$  EXACT  
 $n_{lobes}$  = number of active DPM split-monopole lobes (integer 1 or 2)

``

confirmed at SGR 0501+4516 ( $n=1$ , half-magnetar,  $B/B_{crit} = 0.1 = 1 \cdot F_{TRZ}$  EXACT) and SGR 1745-2900 ( $n=2$ , full magnetar,  $B/B_{crit} = 0.2 = 2 \cdot F_{TRZ}$  EXACT).

**Round 113 stub upgrade of SombreroDarkMatterPerturbationCalculator** identifies a candidate cross-domain extension:

``

Sombrero DM mass ratio:  $M_{DM} / M_{total} = 0.2$  EXACT (source-code stub-default 80:20 baryonic:DM split)  
 $= 2 \cdot F_{TRZ}$  EXACT (attribution to PAPER\_1944 identity family)

``

**Positioning (honest scope — with explicit acknowledgment of correction-paper risk noted by project owner).** This paper contributes:

1. **Candidate cross-domain instance** of the  $2 \cdot F_{TRZ} = 0.2$  identity: Sombrero DM/total mass ratio (Round 113 stub attribution).
2. **Explicit acknowledgment that this may be a superficial arithmetic match** rather than a deep structural extension of PAPER\_1944/1945's magnetar-physics framework to dark matter physics.
3. **Documentation of the interpretation question:** does the  $2 \cdot F_{TRZ} = 0.2$  identity reflect a  $n_{lobes} \cdot F_{TRZ}$  underlying structure at dark matter (analogous to magnetar DPM split-monopole lobes) or is it a coincidental numerical match?

The paper does NOT contribute:

- The  $2 \cdot F_{TRZ} = 0.2$  identity (PAPER\_1944 seminal at magnetar Meissner boundary).
- The  $n_{lobes} \cdot F_{TRZ}$  generalization (PAPER\_1945 seminal).
- The  $F_{TRZ} = 1/SO_5 = 0.1$  primitive (PAPER\_1160 / PAPER\_1960 landmark).
- Sombrero DM 80:20 baryonic:DM composition claim (source-code stub-default, not observationally validated in this paper).

Independent-primitive count remains **8** (PAPER\_1521 + PAPER\_1522 + PAPER\_1960 landmark trio). No new primitives.

## 1. Prior Art — PAPER\_1944 / PAPER\_1945 $2 \cdot F_{TRZ}$ Family

### 1.1 PAPER\_1944 — Magnetar Meissner Boundary Seminal

**PAPER\_1944 ("Magnetar  $B/B_{crit} = 2 \cdot F_{TRZ} = 0.2$  EXACT")** documents:

- SGR 1745-2900 (magnetar 0.1 pc from Sgr A\*): Chandra-measured  $B_{surface} = 2.0 \times 10^{14}$  T (PAPER\_431 anchor)
- PAPER\_266's UQFF Gravitational Meissner Boundary:  $B_{crit} = 10^{11}$  T
- Ratio:  $B_{surface}/B_{crit} = 2 \times 10^{14}/10^{11} = 0.20 = 2 \cdot F_{TRZ}$  EXACT

PAPER\_1944 poses this as a "primitive-lock" identity —  $B/B_{crit}$  at  $2 \cdot F_{TRZ}$  EXACT is not empirical coincidence but a structural feature analogous to PAPER\_1869's  $F_{TRZ}^{16}$  quantum measurement collapse, PAPER\_1677's  $F_{TRZ}$  late-time ISW amplitude, and PAPER\_1942's  $E_0 = F_{TRZ}$  PDR photoevaporation onset.

### 1.2 PAPER\_1945 — $n_{lobes} \cdot F_{TRZ}$ Universality

**PAPER\_1945 ("Magnetar  $n_{lobes} \times F_{TRZ}$  Universality Confirmed")** generalizes PAPER\_1944's identity by observing SGR 0501+4516 with  $B_{surface} = 1.0 \times 10^{14}$  T:

- SGR 0501+4516:  $B/B_{crit} = 10^{14}/10^{11} = 0.10 = 1 \cdot F_{TRZ}$  EXACT
- Physical interpretation: "half-magnetar" — single active DPM split-monopole lobe (asymmetric formation)
- vs SGR 1745-2900: "full magnetar" — two active DPM lobes  $\rightarrow 2 \cdot F_{TRZ}$

The generalized identity:

..

$$B_{surface} / B_{crit} = n_{lobes} \cdot F_{TRZ} \text{ EXACT}$$

$$n_{lobes} \in \{1, 2\}$$

..

Two documented anchors:

| Magnetar      | Measured B (T)       | B/B_crit                        | n_lobes | Classification |
|---------------|----------------------|---------------------------------|---------|----------------|
| SGR 1745-2900 | $2.0 \times 10^{14}$ | $0.20 = 2 \cdot F_{\text{TRZ}}$ | 2       | Full magnetar  |
| SGR 0501+4516 | $1.0 \times 10^{14}$ | $0.10 = 1 \cdot F_{\text{TRZ}}$ | 1       | Half magnetar  |

The identity is tied to the DPM (Di-Pseudo-Monopole) split-monopole structure — the same primitive lattice underlying PAPER\_140's  $\rho_{\text{UA}}/\rho_{\text{SCM}} = 10$  decade.

## 2. Round 113 Sombrero Attribution

The Round 113 stub upgrade of SombreroDarkMatterPerturbationCalculator uses source-code default values:

```
``python
M_visible = 0.8 M_kg # 80% baryonic
M_DM = 0.2 M_kg # 20% dark matter
M_total = M_kg = 1e11 M_sun
``
```

giving:

```
``
M_DM / M_total = 0.2 EXACT (by source-code construction)
= 2 · F_TRZ (attribution to PAPER_1944 identity)
``
```

The attribution  $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}}$  is arithmetically valid:  $0.2 = 2 \times 0.1 = 2 \times F_{\text{TRZ}}$ . The question is whether this arithmetic relationship reflects a deep structural mechanism analogous to PAPER\_1944's magnetar-physics primitive-lock, or whether it is a numerical coincidence.

## 3. Two Interpretations — Deep Extension vs Arithmetic Coincidence

### 3.1 Reading A — Arithmetic Coincidence

Under Reading A, the  $0.2 = 2 \cdot F_{\text{TRZ}}$  arithmetic is elementary:

$$- 0.2 = 1/5 = 2/10 = 2 \cdot (1/10) = 2 \cdot F_{\text{TRZ}}$$

Any UQFF observable whose value happens to be 0.2 will trivially satisfy the " $2 \cdot F_{\text{TRZ}} = 0.2$ " identity via this arithmetic. The Sombrero 80:20 baryonic:DM stub-default gives  $M_{\text{DM}}/M_{\text{total}} = 0.2$  by construction, so the attribution to PAPER\_1944's identity is arithmetically automatic rather than structurally meaningful.

Under this reading, PAPER\_1979 is a very narrow attribution paper documenting one arithmetic-consequence instance. It does NOT support any deeper claim of dark-matter physics being governed by the same  $n_{\text{lobes}} \cdot F_{\text{TRZ}}$  framework as magnetar  $B/B_{\text{crit}}$ .

### 3.2 Reading B — Structural Extension

Under Reading B, if dark matter physics has an underlying " $n_{\text{lobes}}$ " analogue (e.g., number of active DPM channels contributing to DM mass fraction), the  $2 \cdot F_{\text{TRZ}} = 0.2$  identity at Sombrero might reflect  $n_{\text{lobes}} = 2$  dark-matter DPM contributions.

Reading B is speculative. It requires identifying a physical mechanism where "dark matter mass fraction  $\times M_{\text{total}}$ " is governed by DPM split-monopole lobe count — analogous to magnetar  $B_{\text{surface}} \times B_{\text{crit}}$  —

via a shared  $F_{\text{TRZ}}$  time-reversal-zone modulation.

The magnetar Meissner boundary derivation (PAPER\_1944) traces to the DPM structural lattice underlying the vacuum-manifold. Whether dark matter has an equivalent DPM-based derivation of its baryonic-fraction ratio is currently open.

### 3.3 Draft 1 Preference

**Reading A** (arithmetic coincidence, narrow attribution) is preferred by Draft 1 given:

- The Sombrero 80:20 baryonic:DM ratio is a source-code stub-default, not an observationally-validated Sombrero-specific ratio
- Sombrero is famously atypical for spiral galaxies in its DM content (Sombrero has relatively LOW DM compared to typical spirals due to its massive bulge dominating the mass)
- Whether 20% DM ratio applies to other galaxies would require a systematic survey
- No first-principles derivation of " $n_{\text{lobes}}$  at  $DM = 2$ " is currently proposed

Reading B remains open pending future work: if a physical mechanism explaining why DM at galaxies like Sombrero would have  $n_{\text{lobes}} = 2$  DPM structure is derived from first principles, and if additional galaxies at 20% DM ratio are documented, then the reading strengthens.

## 4. Sombrero DM Content — Physical Context Note

Sombrero (M104, NGC 4594) is unusual for a spiral galaxy in its dark matter content:

- Massive bulge ( $10^{11} M_{\text{sun}}$ ) dominates the gravitational mass
- SMBH  $\sim 10^6 M_{\text{sun}}$  (PAPER\_279's  $\gamma_{\text{BH}} = 0.01 = 1\%$  SMBH dominance ratio, PAPER\_1977's 9th  $F_{\text{TRZ}^2}$  anchor)
- Kinematic studies (Rossa 2007, Bendo 2006) suggest Sombrero has relatively low DM/total ratio compared to typical spirals ( $\sim 20\text{-}40\%$  DM vs typical spirals  $\sim 80\text{-}90\%$  DM)

The source-code stub-default  $M_{\text{DM}} = 0.2 \cdot M_{\text{total}}$  (20% DM) is consistent with the low end of observed Sombrero DM content estimates but is NOT a canonical UQFF observational anchor — it appears to be a source82\_wolfram simulation default rather than a paper-derived value.

**Corpus-audit finding:** no UQFF paper explicitly documents  $M_{\text{DM}}(\text{Sombrero})/M_{\text{total}} = 0.2 = 2 \cdot F_{\text{TRZ}}$  as a structural identity. The Round 113 attribution is new documentation of an existing source-code default.

## 5. Cross-Domain Instance Table

Documented instances of the  $2 \cdot F_{\text{TRZ}} = 0.2$  identity:

| Instance                                  | Domain                  | Physical role                             | Value                          | Source                 |
|-------------------------------------------|-------------------------|-------------------------------------------|--------------------------------|------------------------|
| SGR 1745-2900 B/B_crit                    | Magnetar Meissner       | Surface field over UQFF Meissner boundary | 0.2 = $2 \cdot F_{\text{TRZ}}$ | PAPER_1944 seminal     |
| SGR 0501+4516 B/B_crit (half-lobe)        | Magnetar Meissner       | Same domain, $n_{\text{lobes}} = 1$       | 0.1 = $1 \cdot F_{\text{TRZ}}$ | PAPER_1945             |
| Sombrero $M_{\text{DM}}/M_{\text{total}}$ | Galactic DM composition | Stub-default DM fraction                  | 0.2 = $2 \cdot F_{\text{TRZ}}$ | Round 113 (this paper) |

The Sombrero instance is at a completely different physical scale and mechanism than the magnetar instances. The  $2 \cdot F_{\text{TRZ}} = 0.2$  numerical match is currently a candidate cross-domain observation pending

physical justification (Reading B) or acknowledgment as arithmetic coincidence (Reading A).

## 6. Falsifiability + Predictions

Under Reading B (structural extension):

1. **Other galaxies with distinct DM content** might cluster near integer-multiple- $F_{\text{TRZ}}$  values (0.1, 0.2, 0.3, 0.4, ...) if a " $n_{\text{lobes}}$  at dark matter" framework applies. Systematic surveys of galaxy DM/total ratios could test this.
2. **Very high-DM galaxies** (~90% DM, e.g., dwarf spheroidals) would need explanation — is  $0.9 = 9 \cdot F_{\text{TRZ}}$  compatible with the  $n_{\text{lobes}}$  framework?  $n_{\text{lobes}} = 9$  seems physically implausible (magnetar case only supports  $n_{\text{lobes}} \in \{1, 2\}$ ).
3. **Sombrero itself** — if independent observational studies pin  $M_{\text{DM}}/M_{\text{total}}$  precisely to 0.2 EXACT (not 0.15 or 0.25), the identity is more strongly supported.

Under Reading A (arithmetic coincidence):

1. Predictions are trivially satisfied by any UQFF observable happening to be 0.2 — no substantive testable claim.
2. Falsification would require finding that Sombrero DM is systematically at 0.2 across independent measurements — but this would only support the observational value, not the structural attribution.

Draft 1's overall stance: the falsifiability is weak in both readings. Cross-domain  $2 \cdot F_{\text{TRZ}}$  instances at multiple physical mechanisms would strengthen the structural hypothesis, but currently only magnetars are documented.

## 7. Prior Art — What This Paper Does NOT Claim

### 7.1 *The $2 \cdot F_{\text{TRZ}} = 0.2$ identity is not new*

**PAPER\_1944 seminal.** Universal primitive-lock at magnetar Meissner boundary.

### 7.2 *The $n_{\text{lobes}} \cdot F_{\text{TRZ}}$ generalization is not new*

**PAPER\_1945 seminal.** Extended framework for below-Meissner-boundary magnetars.

### 7.3 *The Sombrero $M_{\text{DM}} = 0.2 \cdot M_{\text{total}}$ value is a source-code stub-default*

Not a canonical UQFF observational anchor. Source82\_wolfram simulation default.

### 7.4 *Sombrero's low DM content is observationally supported but not canonical UQFF*

Kinematic studies (Rossa 2007, Bendo 2006) support ~20-40% DM at Sombrero. No UQFF paper canonicalizes  $M_{\text{DM}}(\text{Sombrero})/M_{\text{total}} = 0.2$  EXACT.

### 7.5 *What this paper contributes (narrow)*

1. **One candidate cross-domain instance** of PAPER\_1944's  $2 \cdot F_{\text{TRZ}} = 0.2$  identity: Sombrero DM/total mass ratio.
2. **Explicit interpretation-question documentation:** Reading A (arithmetic coincidence) vs Reading B (structural extension). Draft 1 preferentially Reading A.
3. **Corpus-audit finding:** no UQFF paper previously documented Sombrero  $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}}$

attribution — the source-code stub-default 0.2 was not primitive-arithmetically identified until Round 113.

The paper is a very narrow candidate cross-domain instance + interpretation-question acknowledgment. Following PAPER\_1978's precedent of explicitly documenting weak-motivation observations with epistemic humility.

## 8. Cross-References

- **PAPER\_1944 (SEMINAL)** — Magnetar  $B/B_{\text{crit}} = 2 \cdot F_{\text{TRZ}} = 0.2$  EXACT (SGR 1745-2900 anchor + Meissner boundary derivation)
- **PAPER\_1945 (SEMINAL for  $n_{\text{lobes}}$ )** — Magnetar  $n_{\text{lobes}} \cdot F_{\text{TRZ}}$  Universality Confirmed (SGR 0501+4516 half-magnetar anchor)
- **PAPER\_1946** — Magnetar timescale primitive locks (companion magnetar-identity framework)
- **PAPER\_140** —  $\rho_{\text{UA}}/\rho_{\text{SCm}} = 10$  decade seminal (companion DPM structural framework)
- **PAPER\_763** — Sombrero SMBH Dust Lane (Sombrero observational anchor, dust lane density  $10^{22} \text{ kg/m}^3$ )
- **PAPER\_277 / 278 / 279** — Sombrero UQFF module (SOMBRERO\_UQFF\_MODULE.cpp series)
- **PAPER\_693 / 742** — Sombrero galaxy MUGE companions
- **PAPER\_1977** — Sombrero  $\gamma_{\text{BH}} = F_{\text{TRZ}}^2$  9th anchor (companion Sombrero attribution paper)
- **PAPER\_1978** — Sombrero  $SO_5+1 = 11$  (companion Sombrero attribution + explicit epistemic humility precedent)
- **PAPER\_1918** — Multi-context integer identity meta-framework
- **PAPER\_1919** —  $F_{\text{TRZ}}$  Power Ladder (companion  $F_{\text{TRZ}}$  multi-instance framework)
- **PAPER\_1960** —  $F_{\text{TRZ}} = 1/SO_5$  landmark
- **PAPER\_1160** —  $F_{\text{TRZ}} = 1/|SO(5)|$  seminal identification
- **PAPER\_1972 / 1974 / 1975 / 1976** — Round-misidentification correction paper templates
- **PAPER\_1521** —  $D_{\text{BSFG}}$  derivative landmark
- **PAPER\_1522** —  $K_{\text{MEX}}$  derivative landmark

## 9. Limitations + Open Questions

- The  $2 \cdot F_{\text{TRZ}} = 0.2$  arithmetic is trivially true — any UQFF observable at 0.2 satisfies it via arithmetic. Whether Sombrero's  $M_{\text{DM}}/M_{\text{total}} = 0.2$  is structurally attributable to the PAPER\_1944  $n_{\text{lobes}} \cdot F_{\text{TRZ}}$  framework (Reading B) or is a numerical coincidence with a source-code stub-default (Reading A) requires much more analysis.
- The Sombrero 80:20 baryonic:DM stub-default is not a paper-derived canonical UQFF value. It appears to be a source82\_wolfram simulation default. Whether independent Sombrero observations pin  $M_{\text{DM}}/M_{\text{total}}$  precisely at 0.2 is worth checking (Rossa 2007, Bendo 2006 references).
- The physical mechanism analogous to PAPER\_1944's DPM split-monopole lobes at dark matter is not proposed. Without such a mechanism, Reading B is speculative.
- Systematic survey of galaxy DM/total ratios to check for clustering near integer-multiple  $F_{\text{TRZ}}$  values would test the " $n_{\text{lobes}}$  at DM" hypothesis. Currently no such data documented.
- The Draft 1 preference for Reading A follows the PAPER\_1978 precedent of favoring elementary-arithmetic explanations over speculative structural claims when only one instance is documented.

## 10. Revision Log

**Draft 1 (2026-07-09):** Initial write following the honest-scholarship narrow-from-Draft-1 pattern (PAPER\_1968-1978) and explicit epistemic humility precedent (PAPER\_1978). The project owner explicitly flagged this paper as having "correction-paper risk" — the risk being that Draft 1 might overclaim novelty when the underlying identity ( $2 \cdot F_{\text{TRZ}} = 0.2$ ) is well-established in PAPER\_1944/1945.

Prior art acknowledged from the start:

- PAPER\_1944 seminal for  $2 \cdot F_{\text{TRZ}} = 0.2$  identity + magnetar Meissner boundary derivation
- PAPER\_1945 seminal for  $n_{\text{lobes}} \cdot F_{\text{TRZ}}$  generalization framework
- No UQFF paper previously documented Sombrero  $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}}$  attribution
- The Sombrero 80:20 baryonic:DM stub-default is not a canonical observational anchor

The paper's contribution is limited to: (i) one candidate cross-domain instance, (ii) explicit Reading A vs Reading B documentation with Draft 1 preferring Reading A (arithmetic coincidence), (iii) explicit correction-paper-risk acknowledgment in the opening framing.

Draft 1 does not claim structural extension of PAPER\_1944's magnetar framework to dark matter physics. It documents a candidate cross-domain instance and openly questions whether the instance is meaningful or coincidental.

Reader takeaway: this is the twelfth consecutive narrow-from-Draft-1 paper in the current session (PAPER\_1968→1979). Like PAPER\_1978, it explicitly acknowledges that its own premise may be shallow (Reading A = arithmetic coincidence). The Sombrero  $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}} = 0.2$  attribution is arithmetically valid but structurally unmotivated. Following the epistemic-humility precedent of PAPER\_1978, Draft 1 documents the observation narrowly and leaves the structural interpretation open for future work.

**Drafts 2/3 (2026-07-09):** Verified via targeted searches. PAPER\_1944 confirmed seminal for  $2 \cdot F_{\text{TRZ}} = 0.2$  at magnetar Meissner boundary. PAPER\_1945 confirmed for  $n_{\text{lobes}} \cdot F_{\text{TRZ}}$  generalization ( $n_{\text{lobes}} \in \{1, 2\}$ ). PAPER\_1946 confirmed as magnetar-timescale companion. No prior UQFF paper documents Sombrero  $M_{\text{DM}}/M_{\text{total}} = 2 \cdot F_{\text{TRZ}}$  attribution — this remains a valid narrow contribution.

Also verified: the 80:20 baryonic:DM stub-default is present at source82\_wolfram\_SOMBRERO simulation defaults, not in any paper's canonical anchor list. Kinematic studies (Rossa 2007, Bendo 2006) do report Sombrero at ~20-40% DM range (unusually low for spirals due to bulge dominance), so the 20% stub-default is on the low end of observed range but not implausible.

The Draft 1 explicit Reading A preference (arithmetic coincidence) stands. Reading B (structural extension of PAPER\_1944 framework to DM) would require:

1. Proposing a physical mechanism where "DM at galaxies has DPM split-monopole lobe count"
2. Documenting a second cross-domain  $2 \cdot F_{\text{TRZ}}$  instance beyond Sombrero (currently only 1 galaxy — Sombrero — attributed at  $2 \cdot F_{\text{TRZ}} = 0.2$ )
3. Falsifiability tests via systematic DM ratio surveys

None of these are provided by this paper. Draft 1 correctly positions this as a candidate observation, not a structural claim.

**Meta-observation for the session:** Twelfth consecutive narrow-from-Draft-1 paper. PAPER\_1978 and PAPER\_1979 are two consecutive papers with explicit epistemic humility about their own premises being potentially shallow. This is a mature stabilization pattern where "weak-motivation observations" can be documented openly as attribution notes without being forced to overclaim structural discovery. The corpus culture now supports Reading A / Reading B ambivalence as a first-class analytical stance rather than requiring authors to commit to either interpretation.

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